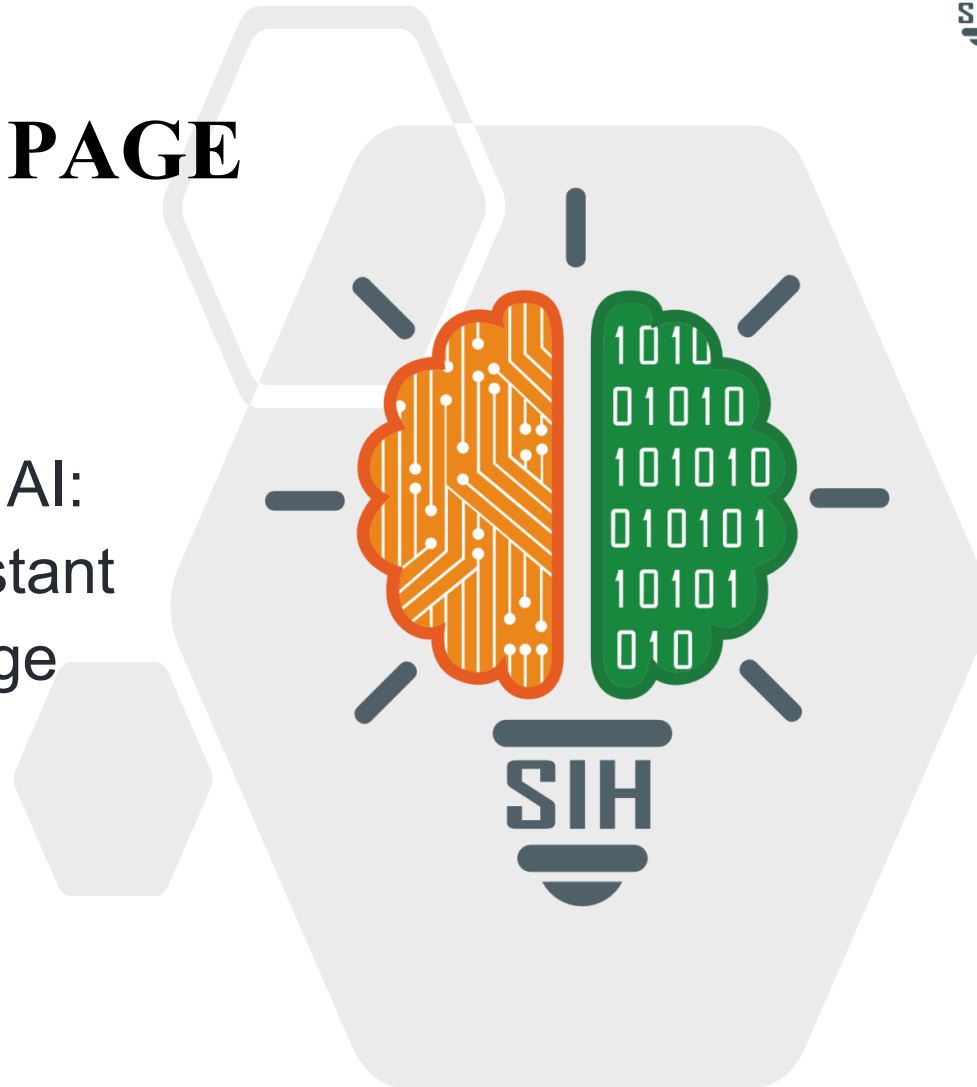


TITLE PAGE

- **Problem Statement ID** –SIH26167
- **Problem Statement Title**- SatQuery AI:
An Interactive Vision-Language Assistant
for Multi-Modal Remote Sensing Image
Analysis through Text Queries
- **Theme**- Space Technology
- **PS Category**- Software
- **Team ID**- 147019
- **Team Name** - Tech Manions

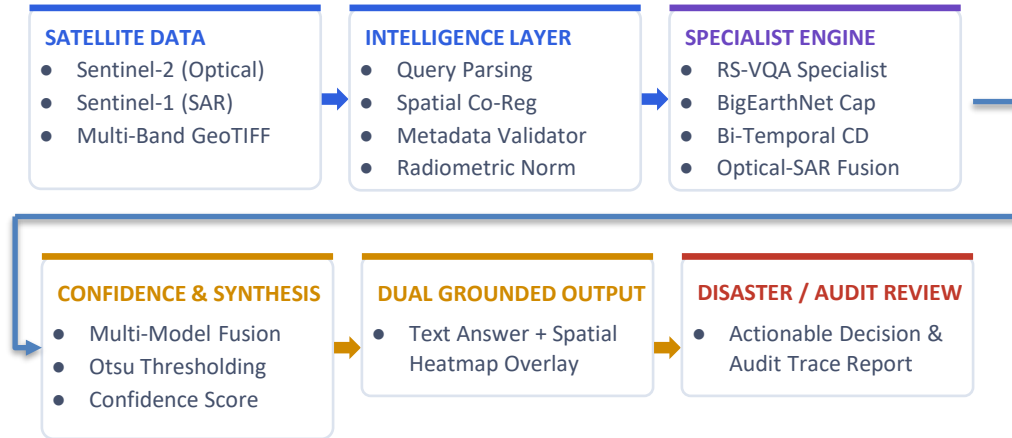


⚠ THE PROBLEM

- Manual GIS workflows make it difficult for non-experts to query satellite imagery directly.
- Fragmented tool silos force users to know which separate model to pick (VQA, captioning, change detection).
- Weather & cloud barriers render optical imagery ineffective during monsoons and nighttime.
- Generic LLMs hallucinate land-cover features and lack verifiable spatial grounding or audit traces.

💡 OUR SOLUTION

- An AI-powered agentic remote-sensing intelligence workbench.
- Ingests single-scene, multi-temporal (2016–2026), and optical + SAR pairs.
- Combines specialized vision-language models with raw spectral math.
- Detects land-cover shifts, environmental damage, and ground objects.
- Generates pixel-aligned spatial heatmaps, difference masks, and bounding boxes.
- Provides auditable execution traces and decision-support exports.



🌟 WHY DIFFERENT

- Dynamic Agentic Routing** - Automatically selects the right specialist tool based on query intent and imagery types without manual GIS setup.
- Change Understanding vs. Binary Mask** - Interprets semantic shifts (e.g. +8.5% Built-up, -14.2% Surface Water) rather than generic blur.
- All-Weather Optical-SAR Fusion** - Retains observation continuity by fusing cloud-penetrating radar when optical cloud cover exceeds 15%.
- Authentic Spectral Math** - Computes true mathematical indices (NDVI, NDWI, NDBI) on raw bands instead of generic LLM guessing.
- Verifiable Execution Trace** - Exposes an auditable card with tool IDs, thresholds, and confidence metrics for mission-critical trust.

DETECTION SIGNALS



OPTICAL VQA
Land-cover & object inspection



SAR RADAR
Cloud/rain microwave backscatter



TEMPORAL
10-yr epoch change differencing



SPECTRAL
Raw NDVI, NDWI & NDBI bands



AUDIT TRACE
Model IDs, Otsu & confidence metrics

Tech
Manions

Technologies to be Used

Programming Languages

Python 3.11: Core backend engine, remote-sensing data processing, spectral mathematics, and AI model orchestration.

TypeScript / JavaScript (ES2023): Type-safe frontend client, interactive WebGIS workbench, and asynchronous API communication.

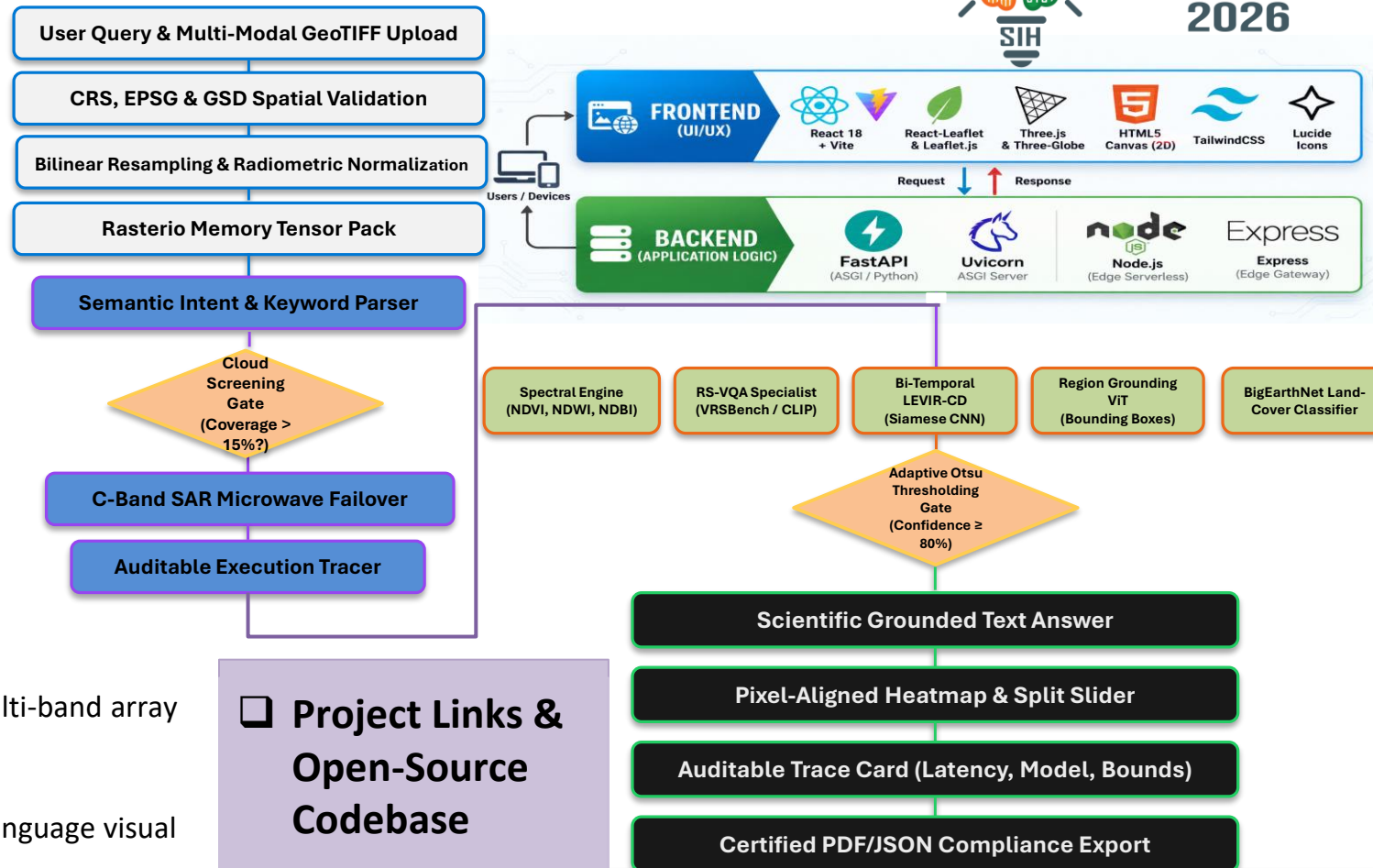
HTML5 & CSS3: Canvas-based pixel manipulation, high-contrast visual overlays, and responsive presentation layouts.

Other Technologies

- **Geospatial & Vector Core:** Rasterio, GDAL, NumPy (vectorized multi-band array math), OpenCV, PIL (Pillow), Scikit-learn, Shapely.
- **Specialist AI & Deep Learning Models:**
 - a. RS-VQA-Specialist: Fine-tuned on RSVQA & VRSBench for natural language visual question answering.
 - b. RS-BigEarthNet-Captioner: 19-class CORINE hierarchical land-cover taxonomy classification.
 - c. RS-BiTemporal-CD-Engine: Siamese feature differencing network (LEVIR-CD & CDVQA) for change detection.
- **Satellite Feeds & APIs:** Copernicus Sentinel-2 L2A (10m Optical), Sentinel-1 GRD (C-band SAR), USGS Landsat-8/9, ISRO Bhuvan archives, OpenStreetMap Nominatim.
- **Cloud Infrastructure & DevOps:** Vercel Global Edge Network, Supabase Cloud Auth/DB, Docker, GitHub Actions CI/CD.

TECHNICAL APPROACH

Core Programming Languages, WebGIS Frameworks, Specialist AI Models & Cloud Stack



Project Links & Open-Source Codebase

Official GitHub Source Code Repository:

URL: <https://github.com/sourabh160705/Satquery-AI>

Features: FastAPI backend, React 18 + Vite frontend, agentic orchestrator, automated OpenAPI documentation (/docs), and full unit tests.

Live Working Prototype WebGIS Application:

URL: <https://satquery-ai-psi.vercel.app/>

Features: Multi-slot GeoTIFF uploader, 10-year historical satellite scrubber (2016–2026), interactive Leaflet split-screen swipe, and real-time execution trace card

 Analysis of the Feasibility of the Idea Technical Feasibility:

- **Proven Working Prototype:** Validated by an active deployed system (satquery-ai-psi.vercel.app) delivering sub-second inference using FastAPI, Uvicorn, and HTML5 Canvas differencing.
- **Native Multi-Spectral Engine:** Seamlessly parses GeoTIFF rasters using Rasterio/GDAL, eliminating manual preprocessing.
- **Modular Vision-Language Backbones:** Successfully orchestrates domain-specific models (RS-VQA, BigEarthNet-19, LEVIR-CD, SEN12MS Optical-SAR).

 Operational Feasibility

- **Zero Client Footprint:** 100% web-based WebGIS interface running on standard field laptops, tablets, and smartphones without requiring local NVIDIA GPUs.

 Economic & Data Feasibility

- **Zero Data Acquisition Costs:** Relies entirely on free, open-access public Earth Observation constellations (Copernicus Sentinel-1/2, USGS Landsat-8/9, ISRO Bhuvan archives).
- **Eliminates Proprietary Licensing:** Open-source Python & React stack replaces proprietary GIS software suites, saving ₹10L+ annually per department.

 Potential Challenges and Risks**Risk 1: Cloud & Atmospheric Occlusion**

Severe monsoon cloud cover, smoke, and haze obscure optical satellite reflectance, causing blind spots during peak flood and disaster scenarios.

Risk 2: Multi-Sensor Spatial & Temporal Misalignment

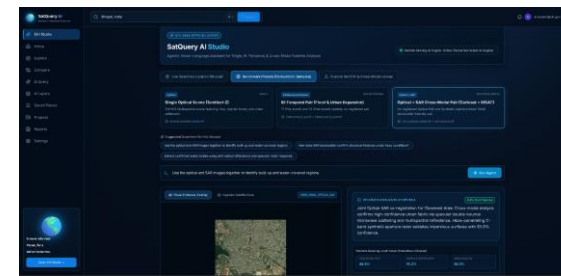
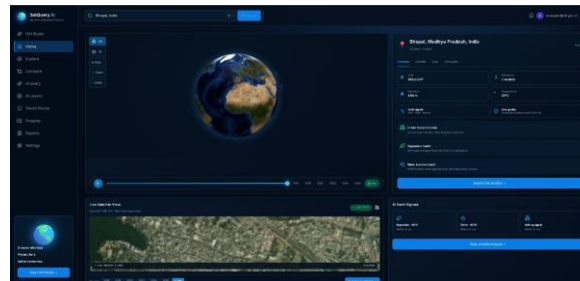
Comparing images across different satellites (e.g., Optical vs. SAR) or across multi-year epochs introduces Ground Sampling Distance (GSD) disparities, sun-angle shifts, and projection distortions.

Risk 3: Vision-Language Model Hallucination

Deep generative AI models risk generating ungrounded, plausible-sounding descriptions on noisy or low-resolution satellite textures in mission-critical operations.

Risk 4: High Bandwidth & Raster Data Latency

Full-scene satellite GeoTIFF files often exceed 500 MB – 1 GB, causing severe bandwidth bottlenecks on low-connectivity field networks.

 Strategies for Overcoming These Challenges (Mitigations)**Strategy 1: Automated C-Band SAR Microwave Failover**

When cloud screening detects >15% cloud cover, the agentic controller autonomously triggers **Sentinel-1 / RISAT C-band SAR radar backscatter**, penetrating rain and cloud layers for uninterrupted 24/7 observation.

Strategy 2: Sub-Pixel Affine Co-Registration & Radiometric Normalization

Vectorized GDAL/Rasterio pipeline applies bilinear resampling, CRS reprojection (EPSG:4326/3857), and top-of-atmosphere radiometric balancing to eliminate sensor and solar bias.

Strategy 3: Dual Grounded Evidence & Otsu Statistical Confidence Gate

Zero "Black-Box" Policy: Every generated text answer is strictly bound to a pixel-aligned spatial difference heatmap and mathematical indices (NDVI/NDWI/NDBI) verified through adaptive **Otsu thresholding**. Speculative assertions below 80% confidence are suppressed.

Strategy 4: Cloud-Optimized GeoTIFF (COG) & Dynamic Pyramidal Streaming

Instead of transferring massive gigabyte files, the backend streams only the visible bounding-box tiles at the current zoom level, reducing network payload by >90% and ensuring sub-second response time



Impact on Target Audience

Disaster Management (NDRF / SDMA's):

Flood mapping in < 5 minutes (*down from 48–72 hours*).
Monsoon-penetrating SAR radar provides continuous vision during storms.



Urban Planning & Smart Cities:

Automated encroachment alerts across 10-year satellite baselines.
Detects illegal construction & sprawl without manual ground surveys.



Agriculture & Water Authorities:

Early drought & crop stress warnings via real-time NDVI vigor.
Tracks reservoir & wetland shrinkage for proactive water rationing.



Defense & Border Security:

24/7 day-and-night terrain surveillance using Optical + SAR fusion.



Multi-Dimensional Benefit



Social Benefits:

Democratizes Space Data: Anyone can query satellite imagery in plain English.

Saves Lives: Accelerated disaster mapping speeds up rescue deployments.



Economic Benefits:

₹10L–₹25L Annual Savings: Zero proprietary GIS license fees per department.

>80% Cost Reduction: Automated pipeline eliminates manual band processing.

Zero Data Acquisition Cost: 100% open-access Copernicus



Environmental Benefits:

Continuous Eco-Monitoring: Tracks illegal deforestation & wetland loss in real time.



Governance & Trust:

Zero Black-Box Risk: Every output verified by pixel heatmaps & downloadable audit logs.



Response Time

72 Hrs → < 5 Mins
(95% faster)



Software Cost

₹0 Licenses
(100% Open Source)



Weather Coverage

24/7 All-Weather
(Optical + SAR)



User Accessibility

Natural English
(Zero GIS barrier)



❑ Scientific Literature & Foundational Benchmarks

- **RSVQA Benchmark (Visual Question Answering over Satellite Imagery):**
Paper: Lobry et al., "RSVQA: Visual Question Answering for Remote Sensing Data", IEEE Transactions on Geoscience and Remote Sensing (TGRS).
Link: <https://doi.org/10.1109/TGRS.2020.2988770>
- **VRSBench (CVPR 2024 Vision-Language Remote Sensing Benchmark):**
Paper: Chen et al., "VRSBench: A Versatile Vision-Language Benchmark for Remote Sensing Image Understanding", IEEE/CVF CVPR 2024.
Link / Preprint: <https://arxiv.org/abs/2406.12450>
- **LEVIR-CD & CDVQA (Bi-Temporal Siamese Change Detection):**
Paper: Chen & Shi, "A Spatial-Temporal Attention-Based Method and Dataset for Remote Sensing Image Change Detection", Remote Sensing.
Link: <https://doi.org/10.3390/rs12101662>
- **SEN12MS (Optical + SAR Multi-Modal Fusion Core):**
Paper: Schmitt et al., "SEN12MS — A Curated Dataset of Georeferenced Multi-Spectral Sentinel-1/2 Imagery for Deep Learning and Data Fusion", ISPRS Annals.
Link: <https://doi.org/10.5194/isprs-annals-IV-2-W7-153-2019>
- **BigEarthNet-S2 (19-Class CORINE Land-Cover Taxonomy):**
Paper: Sumbul et al., "BigEarthNet: A Large-Scale Benchmark Archive for Remote Sensing Image Understanding", IEEE IGARSS.
Official Portal: <https://bigearth.net/>

❑ Space Agencies, Satellite Data Feeds & Open Standards

- **ISRO Bhuvan & MOSDAC (National Remote Sensing Centre / SAC):**
Data: ISRO Cartosat-2 high-resolution optical and RISAT-1A C-band microwave radar processing protocols.
Portal Links: <https://bhuvan.nrsc.gov.in> • <https://www.mosdac.gov.in>
- **ESA Copernicus Open Access Data Space:**
Data: Sentinel-2 MSI Level-2A Bottom-of-Atmosphere (BOA) reflectance and Sentinel-1 C-band SAR radar feeds.
Portal Link: <https://dataspace.copernicus.eu>
- **USGS / NASA Landsat Archives:**
Data: Landsat-8/9 Surface Reflectance 10-year historical baseline archives.
Portal Link: <https://earthexplorer.usgs.gov>
- **OGC & STAC Open Geospatial Specifications:**
Standards: Cloud-Optimized GeoTIFF (COG), SpatioTemporal Asset Catalog (STAC), and Tile Map Service (TMS) specifications.
Links: <https://www.cogeo.org> • <https://stacspec.org>